GAMSA · Gender-Affirming Medicines South Australia

Ventrogluteal intramuscular administration of testosterone undecanoate 1000 mg/4 mL

Standard operating procedure and training guide

VERSION	EFFECTIVE	REVISED	NEXT REVIEW
2.1	5 October 2025	16 October 2026	October 2027, or on any change to the Product Information

Current version

Prepared by Theodore South, BPharm (Hons), registered pharmacist, Adelaide. A template for adaptation to local policy, offered freely; not endorsed by any health service. Clinics remain responsible for their own procedures, credentialling and emergency arrangements. Where this document and the Product Information differ, the Product Information prevails [1].

At a glance

Site: the gluteal muscle, the only site for which the manufacturer has clinical experience [1][3]; this procedure uses the ventrogluteal site (gluteus medius), SA Health's preferred site for long-acting injections [4].

Needle: draw up with 18G; inject with 21G × 38 mm, or 21G × 50 mm where BMI is over 30 (20–22G acceptable) [5][10].

Technique: 90° to the skin, aspirate, inject the full 4 mL very slowly over approximately 2 minutes, no massage afterwards [1][5][10].

Observe: during and immediately after the injection, and for at least 30 minutes under this procedure, because POME typically presents within 30 minutes [12]; the Product Information gives no fixed time [1][3].

Emergencies: POME — stop, sit up, oxygen, monitor, 000 if not settling. Anaphylaxis — lie flat, adrenaline 1:1000 IM 0.01 mg/kg up to 0.5 mg, 000 [7][8].

Screen before each injection: current chest infection or breathlessness, previous POME or reaction to an oily injection, bleeding disorder or anticoagulant use (deep IM injection not advisable; seek haematology advice or use a non-injectable product) [1][11]. A pregnancy test before each injection is recommended for anyone who could become pregnant [1][12].

Interval: maintenance every 10 to 14 weeks; the first-to-second interval may be shortened to a minimum of 6 weeks [1][3].

Package contents: this SOP and training guide · treatment-room card (one page) · competency checklist · adverse event form · clinic audit tool. All at https://gamsa.au/medicines/testosterone-undecanoate.

Section 1: Standard operating procedure

Purpose and scope

To ensure the safe, consistent administration of testosterone undecanoate 1000 mg/4 mL (Reandron 1000 and equivalent brands) by deep intramuscular injection into the gluteal muscle, with the ventrogluteal site preferred, minimising the risk of injury and of pulmonary oil microembolism (POME) [1][4][5][6]. Applies to credentialled clinicians (medical practitioners, nurse practitioners and registered nurses) who administer this product in the facility. It does not replace the Product Information [1].

Responsibilities

- **Prescribing clinician:** confirms the indication, the dose schedule and informed consent [1].
- **Administering clinician:** trained and assessed as competent in ventrogluteal intramuscular technique and in recognising and managing POME and anaphylaxis (Section 2); competency recorded on the staff file [5].
- **Clinic manager:** ensures emergency equipment (oxygen, adrenaline, monitoring, a means of calling 000) is available and in date, and keeps the current Product Information on file [1][8].

Pre-procedure checks

- Verify patient identity, indication, allergies and informed consent, including any previous reaction to an oil-based injection [1].
- **Screen:** current chest infection or breathlessness (defer if unwell, so that POME is not masked or mistaken); any previous POME or reaction to an oily injection; any bleeding or coagulation disorder or anticoagulant use. The Product Information advises against deep intramuscular injection in those with a bleeding or coagulation disorder, including anticoagulant users, because of the risk of haematoma: use a non-injectable testosterone product or seek a haematologist's advice [1][11].
- **Pregnancy test (recommended):** before each injection for anyone who could become pregnant; testosterone is contraindicated in pregnancy (Product Information sections 4.3 and 4.6) [1][12].
- Confirm the patient can remain for the observation period and has a way home.
- Confirm the product, dose (1000 mg in 4 mL), expiry and batch number. Check the Product Information for contraindications and precautions [1][2].

Equipment

- testosterone undecanoate 1000 mg/4 mL (Reandron 1000 and equivalent brands), one 4 mL vial or ampoule [1].
- 18G needle for drawing up: the solution is a viscous oil [5][10].
- Injection needle 20–22G [10]; this procedure specifies 21G × 38 mm, or 21G × 50 mm where the BMI is over 30, so that the needle reaches muscle [5].
- 5 mL syringe, gloves, antiseptic swabs, gauze, adhesive dressing, sharps container [5].
- Emergency kit: oxygen, adrenaline for intramuscular use, monitoring equipment, a means of calling 000; ASCIA flowchart displayed [8].

Procedure

1	**Position the patient** side-lying with the upper hip exposed and relaxed, or supine with slight hip flexion. Ensure comfort and privacy [5].
2	**Landmark the ventrogluteal site** with the hand opposite to the hip (left hand for the right hip): heel of the hand over the greater trochanter, index finger towards the anterior superior iliac spine, middle finger spread back along the iliac crest. The centre of the V between the two fingers, over gluteus medius, is the injection point (Appendix A) [4][5].
3	**Prepare the site** with antiseptic and allow it to dry. Put on gloves [5].
4	**Draw up** the full 4 mL with the 18G needle, change to the injection needle and expel air [1][5][10].
5	**Insert the needle at 90°** to the skin, to a depth that places the tip in muscle, not subcutaneous tissue. Aspirate: if blood appears, withdraw, replace the needle and choose a new site [5][10].
6	**Inject very slowly** over approximately 2 minutes [10]. The Product Information requires very slow injection without giving a time [1][3]; the manufacturer's administration guidance for the same formulation specifies about 2 minutes [10]. Slow injection reduces the pressure that forces oil into vessels [6][7].
7	**Withdraw the needle** and apply gentle pressure with gauze. Do not massage the site. Apply a dressing and dispose of sharps immediately [5].
8	**Observe** during and immediately after the injection [1][3], and for at least 30 minutes under this procedure: POME typically presents within 30 minutes [12], and the Product Information gives no fixed time. Book the next appointment before the patient leaves.

Observation and management of reactions

Pulmonary oil microembolism presents within minutes as cough or an urge to cough, breathlessness, malaise, sweating, chest pain, dizziness, paraesthesia or syncope [1][3]. It is uncommon: 28 cases classified as POME in 90,092 doses of the US 750 mg/3 mL product distributed over 4.3 years (2014–2018), a reported rate below 0.1% per injection, against 0.8–2.1% in clinical trials where every injection was observed [6]. Symptoms usually resolve within 30 minutes to a few hours, with supportive care if needed [6][12]. Anaphylaxis is rare but possible [1][3].

Suspected POME

Stop the injection if still in progress. Reassure and sit the patient up (legs outstretched) if breathless [7][8].

High-flow oxygen; monitor pulse, blood pressure, oxygen saturation and respiratory rate [7].

Call 000 immediately for severe breathing difficulty, collapse or signs of anaphylaxis [7][8]. Mild POME symptoms usually settle within 30 minutes to a few hours with supportive care [6][12].

Document, notify the prescriber, complete the adverse event form (Appendix C) and report to the TGA [9].

Suspected anaphylaxis (ASCIA, July 2026)

Lay the patient flat with legs slightly raised; do not let them stand or walk. If breathing is the main problem they may sit with legs outstretched [8].

Adrenaline 1:1000 intramuscularly into the outer mid-thigh: 0.01 mg/kg, up to 0.5 mg (0.5 mL) per dose for an adult. Call 000. Repeat after 5 minutes if there is no improvement [8].

Oxygen, monitoring, and observation in hospital for at least 4 hours after the last dose of adrenaline [8].

Document, notify the prescriber, complete the adverse event form and report to the TGA [9].

Documentation

Record the date and time, product, batch number and expiry, dose, side injected (for example “right ventrogluteal”), needle gauge and length, the administering clinician, any reaction, the patient's condition at the end of the observation period, and the next due date. Maintenance injections are usually given every 10 to 14 weeks; the interval between the first and second injection may be shortened to a minimum of 6 weeks [1][3].

Reporting, escalation and review

Any serious or unexpected reaction, especially POME or anaphylaxis, is escalated immediately to the on-call medical officer, recorded in the local incident system and reported to the TGA through AEMS [9]. The event is reviewed, and this procedure or the training updated if needed. The procedure is reviewed annually, or whenever the Product Information, national guidance or local policy changes.

Section 2: Training guide

Learning objectives

On completion the clinician can:

- explain why the gluteal muscle, and the ventrogluteal site in particular, is used for this large-volume oily injection [1][4][5];
- landmark the ventrogluteal site accurately [4][5];
- choose needle gauge and length for the patient's build [5][10];
- inject 4 mL of oily solution deep into muscle over approximately 2 minutes [1][3][10];
- recognise and manage POME and anaphylaxis [6][7][8];
- document and report correctly [9].

Rationale

Reandron 1000 is licensed for deep intramuscular injection into the gluteal muscle, the only site for which clinical experience has been obtained [1][3]. SA Health has moved long-acting injections to the ventrogluteal site because it is safer and absorbs medicine well [4]: it has substantial muscle bulk and lies away from the sciatic nerve and the superior gluteal vessels, which the dorsogluteal site risks [4][5]. The ventrogluteal site accepts the 4 mL volume [5].

Anatomy and landmarking

The site overlies gluteus medius and minimus, bounded by the greater trochanter (lateral), the anterior superior iliac spine (anterior) and the iliac crest (posterior). Landmark as in Section 1 and Appendix A [4][5]. Where landmarks are hard to feel, take time, ask a colleague to confirm, and choose the longer needle so the injection reaches muscle [5].

Needle and equipment selection

Draw up with an 18G needle, as the oil is viscous. Inject with a 20–22G needle [10]; this procedure specifies 21G, 38 mm for most adults and 50 mm where the BMI is over 30 [5]. A wider lumen lets the oil flow with less pressure.

Injection technique

Follow Section 1. The points most often done poorly are injecting too fast (the target is about 2 minutes for the 4 mL [10]), injecting too shallow, and massaging the site afterwards; each increases the chance of oil entering the circulation or leaking back along the needle track [5][6][7].

Recognising and managing complications

- **POME:** sudden cough, urge to cough, breathlessness, chest tightness, sweating, dizziness, paraesthesia or syncope within minutes of injection. Manage as in Section 1. Most episodes are self-limiting; escalate if severe [1][6][7].
- **Anaphylaxis:** manage per ASCIA: lie flat, adrenaline 1:1000 IM outer mid-thigh 0.01 mg/kg up to 0.5 mg (adult), call 000, repeat after 5 minutes if needed, observe at least 4 hours after the last dose [8].
- **Local:** pain, haematoma, oil nodule, infection; manage per local wound care and escalate if infection is suspected [5].

Practical training and competency

- A short teaching session covering the Product Information, POME and anaphylaxis (30–45 minutes) [1][6][7][8].
- Demonstration of landmarking and injection on a model (15–20 minutes) [4][5].
- At least three supervised ventrogluteal injections, documented [5].
- Direct observation against the competency checklist (Appendix B), and a short quiz on recognising POME and anaphylaxis.

Patient information and counselling

- Explain the schedule: usually every 10 to 14 weeks once established [1][3].
- Explain that the injection goes into the side of the hip, takes about two minutes to give [10], and is followed by at least 30 minutes of observation under this procedure (Section 1, step 8).
- Ask the patient to report sudden cough, breathlessness, dizziness, faintness or chest symptoms immediately, during the visit or after it [1][3].
- Give the Consumer Medicine Information leaflet [1]; the plain-language appointment guide below and the Easy Read guide at gamsa.au/easy/injections can be printed for patients.

At the appointment, step by step

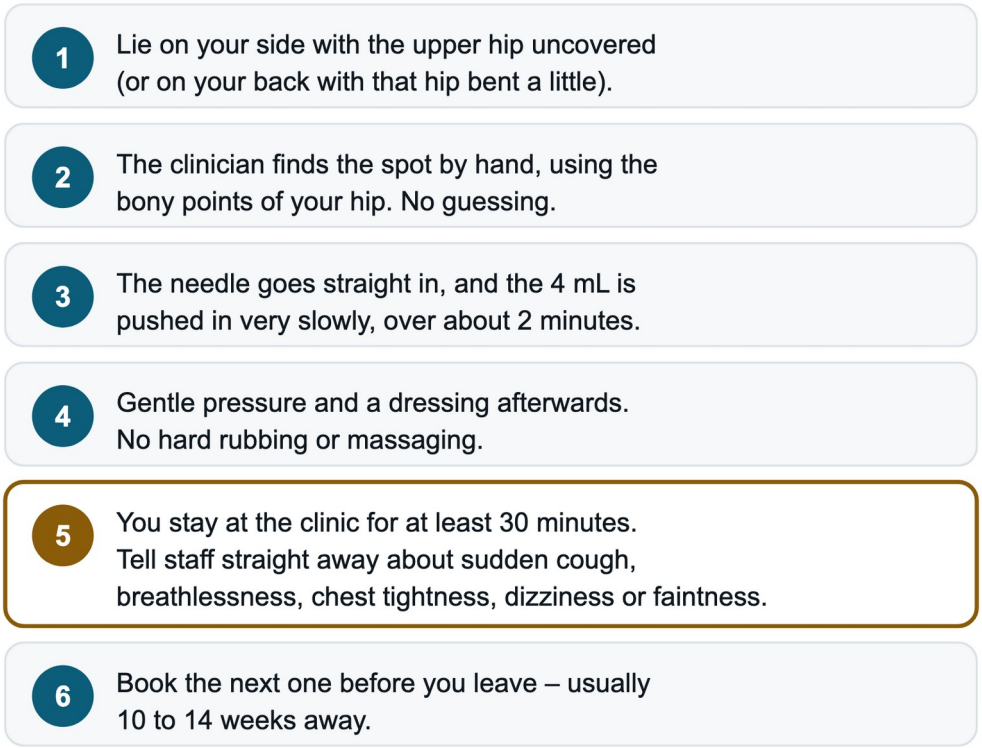


Figure 1. The appointment, step by step (gamsa.au). Suitable to print for patients.

Appendix A: Landmarking the ventrogluteal site

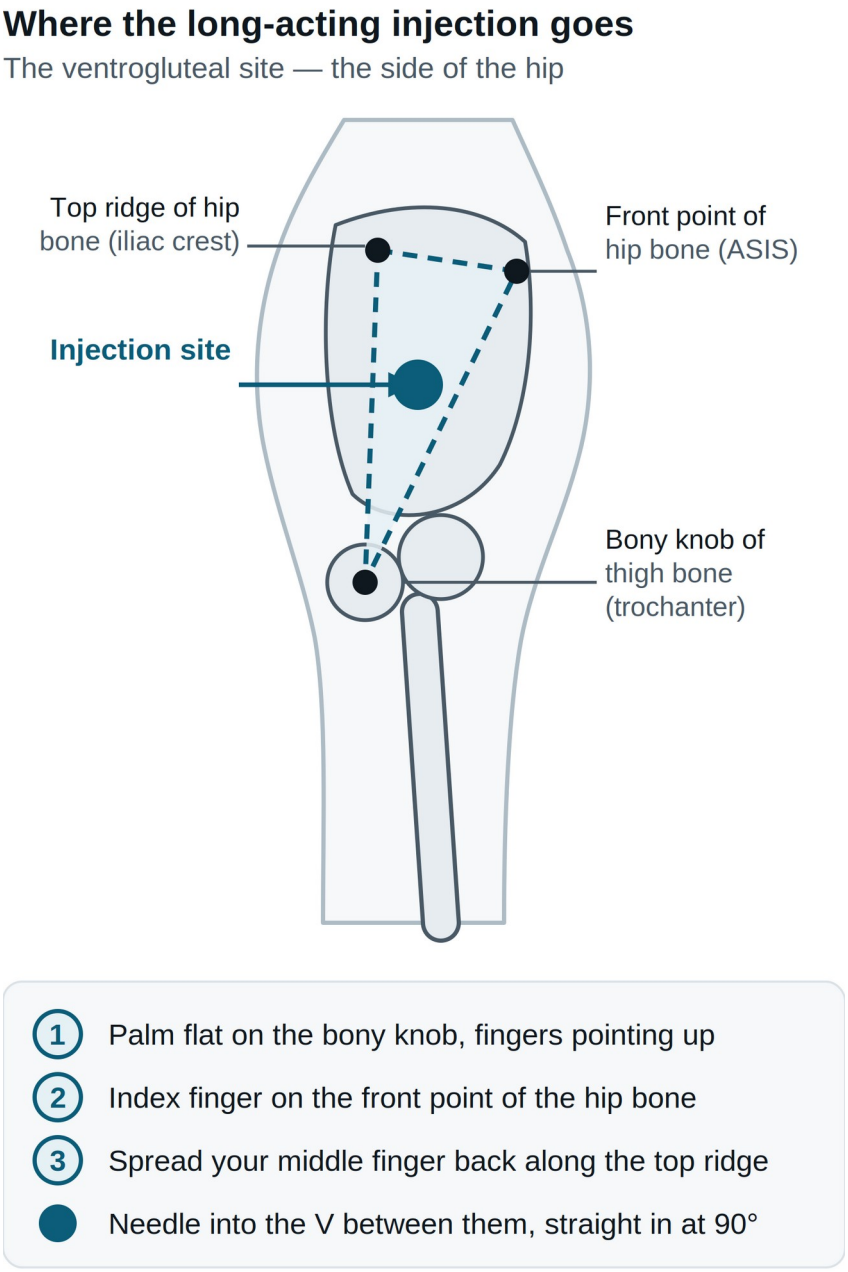


Figure 2. The ventrogluteal site and its three bony landmarks (gamsa.au), drawn from the SA Health brochure [4] and SESLHD guideline [5].

Appendix B: Competency assessment checklist, and Appendix C: Adverse event reporting and escalation form, are separate documents in this package.

References

[1] Aspen Pharmacare Australia. Australian Product Information: Reandron 1000 (testosterone undecanoate) solution for injection, PI version 2.0, date of revision 11 December 2025. Canberra: TGA eBS. Accessed 16 October 2026. https://www.ebs.tga.gov.au/ebs/picmi/picmirepository.nsf/pdf?OpenAgent&id=CP-2024-PI-02502-1

[2] Therapeutic Goods Administration. REANDRON 1000 testosterone undecanoate 1000 mg/4 mL injection vial, ARTG 213942. Canberra: TGA. Accessed 16 October 2026. https://www.tga.gov.au/resources/artg/213942

[3] Bayer New Zealand. Reandron 1000 (1000 mg/4 mL solution for injection) New Zealand Data Sheet, revised 23 December 2025. Wellington: Medsafe. Accessed 16 October 2026. https://medsafe.govt.nz/profs/Datasheet/r/Reandron1000inj.pdf

[4] SA Health. Changes to your long acting injection: what you and your support team need to know (ventrogluteal site patient brochure). Adelaide: Government of South Australia. Accessed 16 October 2026.

https://www.sahealth.sa.gov.au/wps/wcm/connect/e185e690-bc60-402f-b264-5ed189fd0390/Long+Acting+Intramuscular+Injections+Brochure+(1).pdf?MOD=AJPERES

[5] South Eastern Sydney Local Health District. Intramuscular injection in mental health. Guideline SESLHDGL/076, version 2.6, 20 January 2025. Sydney: NSW Health. The public copy is no longer online (checked 16 October 2026); SESLHD lists its mental health policies for staff access only: https://www.seslhd.health.nsw.gov.au/policies-and-publications/functional-group/89

[6] Pastuszak AW, Hu Y, Freid JD. Occurrence of pulmonary oil microembolism after testosterone undecanoate injection: a postmarketing safety analysis. Sex Med. 2020;8(2):237-242. doi:10.1016/j.esxm.2020.01.009 https://doi.org/10.1016/j.esxm.2020.01.009

[7] Adam JA, Pastuszak AW, Christensen MB, et al. Occurrence of pulmonary oil microembolism (POME) with intramuscular testosterone undecanoate injection: literature review. Int J Impot Res. 2023;35(5):439-446. doi:10.1038/s41443-022-00585-1 https://doi.org/10.1038/s41443-022-00585-1

[8] Australasian Society of Clinical Immunology and Allergy. ASCIA guidelines: acute management of anaphylaxis. Updated July 2026. Accessed 16 October 2026. https://www.allergy.org.au/images/ASCIA_HP_Guidelines_Acute_Management_Anaphylaxis_July_2026.pdf

[9] Therapeutic Goods Administration. Report an adverse event or safety problem; Adverse Event Management System (AEMS) online reporting. Canberra: TGA. Accessed 16 October 2026. https://aems.tga.gov.au/

[10] Laboratorios Liconsa. Testosterone undecanoate 1000 mg/4 mL: guidance on administration (risk minimisation material approved by the MHRA), version 2.0, July 2024. Same formulation as Reandron 1000. Accessed 16 October 2026. https://www.medicines.org.uk/emc/rmm/100105/Document

[11] ADVANZ PHARMA (Australia). Australian Product Information: Testosterone ADVZ 1000 (testosterone undecanoate) solution for injection. Canberra: TGA eBS. Accessed 16 October 2026. (Generic brands Gonadron and Rejunon 1000 are equivalent; check each brand's own PI.) https://www.ebs.tga.gov.au/ebs/picmi/picmirepository.nsf/pdf?OpenAgent&id=CP-2023-PI-02713-1

[12] Australian Professional Association for Trans Health (AusPATH). AusPATH Australian Standards of Care for Informed Consent Gender-Affirming Hormone Therapy (2026), final version 30 April 2026, pp 56 and 58. Accessed 16 October 2026. https://auspath.org.au/wp-content/uploads/2026/05/Standards_of_Care_Digital_Final_30_April_2026.pdf

Resources

- Patient-facing explanation of the appointment, with the landmarking and appointment infographics (print for patients) https://gamsa.au/using#sec-using-3
- Medicine page: testosterone undecanoate on GAMSA (doses, PBS listing, supply status, this package) https://gamsa.au/medicines/testosterone-undecanoate
- Easy Read guide: Having an injection https://gamsa.au/easy/injections
- ASCIA anaphylaxis action plans and flowcharts (print for the treatment room) https://www.allergy.org.au/hp/anaphylaxis
- Consumer Medicine Information for Reandron 1000 (give to patients) https://www.safetyandquality.gov.au/medicine-finder/reandron-1000-solution-for-injection
- TGA adverse event reporting (AEMS) https://aems.tga.gov.au/

Local adoption record

Adopted or adapted by (service)	Approved by	Date